

In the upcoming formulas, vectors are expressed in bold ($\mathbf{v}$), a vector with a hat implies that it is a unit vector ($\hat{\mathbf{x}}$) and quantities that are in reduced units are starred (v^*).

Table I
REDUCED (DIMENSIONLESS) UNITS

Property	Symbol	Reduced form
Length	l^*	l/σ
Time	t^*	$\frac{D_T}{\sigma^2} t = \frac{t}{\tau}$
Magnetic dipole	m^*	m/μ
Magnetic Force	$\mathbf{F}^*$	$\frac{2\pi\sigma^4}{3\mu_0\mu^2} \mathbf{F}$
White noise	ξ^*	$\sqrt{\tau}\xi$

Where σ is the characteristic length which, in this case, is equal to the diameter of the particle; τ_R is the characteristic time which is the time where a particle maintains its orientation; μ is the characteristic dipole moment; and μ_0 is the magnetic permeability of vacuum.

Langevin equation (dimensionless)

The equation used to describe the trajectory Active Brownian Particles (ABPs) is

$$\begin{cases} \frac{d\mathbf{r}_i}{dt} = v_0 \hat{\mathbf{e}}_i + \frac{1}{\gamma_T} \mathbf{F}_i + \sqrt{2D_T} \xi_i^T \\ \frac{d\theta_i}{dt} = \sqrt{2D_R} \xi_i^R \end{cases}, \quad (1)$$

where $\mathbf{r}_i$ is the position vector of the particle i , t is time, v_0 is the speed due to the activity, $\hat{\mathbf{e}}_i$ is a unit vector whose direction is determined by θ , $\mathbf{F}_i$ is the force due to the dipole-dipole interactions between the particle i and the rest of the particles in the system, γ_T is the translational drag coefficient, D_T and D_R are the translational and rotational diffusion coefficient, respectively, and ξ_i^j are white noises following a Gaussian distribution with zero mean and unit variance. The relevant reduced units are shown in Table I.

We start with the spatial component, which can be rewritten as follows

$$\frac{d\mathbf{r}_i}{dt} = v_0 \hat{\mathbf{e}}_i + \frac{-\nabla U_{mag}(\mathbf{r})}{\gamma_T} + \sqrt{2D_T} \xi_i, \quad (2)$$

Where $U_{mag}(\mathbf{r})$ is the magnetic energy.

Substituting the reduced units into equation (1),

$$\frac{\sigma}{\tau} \frac{d\mathbf{r}_i^*}{dt^*} = v_0 \hat{\mathbf{e}} - \frac{1}{\sigma} \frac{\nabla^* U_{mag}(\mathbf{r})}{\gamma_T} + \sqrt{\frac{2D_T}{\tau}} \xi_i^* \quad (3)$$

Rearranging the equation, we have

$$\frac{d\mathbf{r}_i^*}{dt^*} = \frac{v_0 \tau}{\sigma} \hat{\mathbf{e}} - \frac{\tau}{\sigma^2} \frac{\nabla^* U_{mag}(\mathbf{r})}{\gamma_T} + \sqrt{\frac{2D_T \tau}{\sigma^2}} \xi_i^* \quad (4)$$

Now, from the different constant, it is possible to identify various dimensionless numbers. Starting from the activity component,

$$\frac{v_0 \tau}{\sigma} = \frac{v_0 \sigma}{D_T} = Pe, \quad (5)$$

Where we have used $\tau = \sigma^2/D_T$, we see that it contains the Peclet number, which compares the activity and the diffusion. Next, in the magnetic term, consider that

$$U_{mag} = -\frac{\mu_0}{2\pi r^3} \left[1 - \frac{3}{2} \sin^2 \theta \right] = -U_{max} \frac{\sigma^3}{r^3} \left[1 - \frac{3}{2} \sin^2 \theta \right] = -U_{max} u, \quad (6)$$

Where U_{max} is the maximum value of magnetic energy, we have

$$\frac{\tau}{\sigma^2} \frac{U_{max}}{\gamma_T} = \frac{\tau}{\sigma^2} \frac{U_{max}}{k_B T} D_T = \Gamma \frac{D_T \tau}{\sigma^2} = \Gamma \quad (7)$$

Where we find the dimensionless number Γ , which compares magnetic energy with thermal energy. Finally, in the noise term, we have

$$\sqrt{\frac{2D_T \tau}{\sigma^2}} = \sqrt{2} \quad (8)$$

The resulting equation then becomes

$$\frac{d\mathbf{r}^*}{dt} = Pe \hat{\mathbf{e}} - \frac{\Gamma}{2} \nabla^* u(\mathbf{r}) + \sqrt{2} \xi_i^* \quad (9)$$

Now we proceed with the angular component

$$\frac{d\theta_i}{dt} = \sqrt{2D_R} \xi_i^R \quad (10)$$

Substituting the reduced units

$$\frac{1}{\tau} \frac{d\theta_i}{dt^*} = \sqrt{\frac{2D_R}{\tau}} \xi_i^{R*} \quad (11)$$

We can rearrange the equation

$$\frac{d\theta_i}{dt^*} = \sqrt{\frac{2\tau}{\tau_R}} \xi_i^{R*} \quad (12)$$

We see that there appears a dimensionless number which relates the translational relaxation time τ with the rotational relaxation time τ_R .

Dimensionless number Le

In this part, the objective is to adapt Le so it is possible to obtain its value using the reduced units implemented in the simulation. During the simulation, we have information about F_{act}^* , T^* , m^* , γ_T^* and γ_R^* . Thus, we ought to leave Le as a function of the mentioned variables.

Starting from the definition of Le ,

$$Le = \frac{W_{act}}{U_{mag}} \quad (13)$$

where W_{act} is the work done by the force of the activity (F_{act}) and U_{mag} is the maximum magnetic interaction energy. We first work out W_{act} ,

$$W_{act} = \gamma_T v \sigma = F_{act} \sigma \quad (14)$$

where we have used $F_{act} = \gamma_T v$. Now, we substitute the corresponding dimensionless values,

$$W_{act} = \frac{\epsilon}{\sigma} F_{act}^* \sigma = \epsilon F_{act}^* \quad (15)$$

where ϵ is the characteristic energy, in this case $\epsilon = \frac{\mu_0 \mu^2}{4\pi \sigma^3}$, which is the characteristic magnetic energy. Now, we work out U_{mag} , where we take its maximum value,

$$U_{mag} = \frac{\mu_0 m^2}{2\pi \sigma^3} = \frac{\mu_0 \mu^2}{2\pi \sigma^3} m^{*2} = 2\epsilon m^{*2} \quad (16)$$

Finally

$$Le = \frac{W_{act}}{U_{mag}} = \frac{\epsilon F_{act}^*}{\epsilon m^{*2}} = \frac{F_{act}^*}{2m^{*2}} \quad (17)$$

Dimensionless number Pe

Just like the previous section, here we aim to adapt the Peclet number so that it can be calculated using the reduced units implemented for the simulations. The Peclet number is given by

$$Pe = \frac{v_0 \sigma}{D_T} \quad (18)$$

Considering that $F_{act} = \gamma_T v_0$ and $D_T = k_B T / \gamma_T$, we have

$$Pe = \frac{F_{act} \sigma}{k_B T} \quad (19)$$

Meaning that it is possible to interpret the Peclet number as the ratio between the work done by the activity and the thermal energy. Now, we substitute the corresponding dimensionless values,

$$Pe = \frac{\epsilon F_{act}^* \sigma}{\sigma \epsilon T^*} = \frac{F_{act}^*}{T^*} \quad (20)$$

Dimensionless number Γ

The expression for Γ is

$$\Gamma = \frac{U_{mag}}{k_B T} \quad (21)$$

Using the result obtained in equation (9), we have the following

$$\Gamma = \frac{2m^{*2}}{T^*} \quad (22)$$

Note that now, it is possible to express Le as

$$Le = \frac{Pe}{\Gamma} \quad (23)$$

Distance of no return

Consider a particle at one end of the chain, we assume that if the active force separates said particle with its closest neighbor by a distance $r_{no\ return}$, where $U_{mag}(r_{no\ return}) = k_B T$, then the particle diffuses away from the chain such that the attractive magnetic force from its former neighbors is unable to recover this particle, i.e. it leaves the chain. To estimate $r_{no\ return}$, we consider

$$U_{mag}(r_{no\ return}, \theta) = \frac{\mu_0 m^2}{2\pi r_{no\ return}^3} \left(1 - \frac{3}{2} \sin^2 \theta\right) = k_B T \quad (24)$$

The maximum value of U_{mag} is achieved when $\theta = 0$, then, solving for $r_{no\ return}$ when $\theta = 0$ we obtain the distance from where we are sure that the particle is separated,

$$\frac{\mu_0 m^2}{2\pi r_{no\ return}^3} = k_B T, \quad (25)$$

$$r_{no\ return} = \sqrt[3]{\frac{\mu_0 m^2}{2\pi k_B T}} = \sqrt[3]{\Gamma \sigma^3}. \quad (26)$$

Now, suppose that the active force is known, the time needed to reach such distance can then be estimated

$$\tau_{no\ return} = \frac{r_{nr}}{v_0} = \frac{r_{nr} \gamma_T}{F_{act}} = \sqrt[3]{\Gamma \sigma^3} \frac{\gamma_T}{F_{act}} \quad (27)$$

In reduced units

$$\tau \tau_{nr}^* = \sqrt[3]{\Gamma \sigma^3} \frac{\epsilon \tau}{\sigma^2} \frac{\sigma}{\epsilon} \frac{\gamma_T^*}{F_{act}^*} \quad (28)$$

$$\tau_{nr}^* = \sqrt[3]{\Gamma \sigma^3} \frac{\gamma_T^*}{F_{act}^*} \quad (29)$$